

# What this project does — a plain-language explanation

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*Qiskit Hackathon 2026, Topic 5: "In the Shadow of the Hadamard Test" (based on Faehrmann, Eisert & Kueng, PRL 135, 150603, 2025).*

*Every number quoted below is traceable to a row in RESULTS.md, which records the command or notebook cell that produced it.*

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## The one-sentence version

We ran a standard quantum-computing experiment that normally throws away most of its measurement data, kept the "garbage" instead, and showed that this garbage contains enough extra information to label the energy levels of a small magnet — with no extra circuits, at a measured cost of about 7% more statistical noise.

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## The longer version

### 1. The setup: a tiny quantum magnet

Our test subject is a chain of three quantum spins — think of three tiny magnets in a row that can each point up or down, and that push and pull on each other. Simulating how such a system evolves over time is one of the core things quantum computers are expected to be good at.

Some facts about this little system are known exactly (a laptop can compute them by brute force, because three spins are small). That is the point: we always have an exact answer sheet to grade ourselves against. The rule we imposed on ourselves all week: **an estimate only counts if it agrees with the exact answer within its own error bars, and every claim traces back to a logged run.**

### 2. The standard tool: the Hadamard test

The "Hadamard test" is a classic quantum circuit. You add one extra helper qubit (an *ancilla*) to your system, entangle it cleverly with the time evolution, and then measure it. The ancilla's answers — just  $+1$ s and  $-1$ s — average out to a number physicists care about: the *signal*  $\chi(t)$ , which encodes the energy spectrum of the system the way a chord encodes the notes inside it.

Here's the wasteful part: in the standard protocol, **you only read the helper qubit. The three system qubits get measured too, but their results are treated as garbage and discarded.**

### 3. The paper's idea: read the garbage

The 2025 paper this challenge is built on makes a simple, clever observation: if you measure those three "garbage" qubits in randomly chosen directions (a technique called *classical shadows*), the discarded data is not garbage at all. From the **same** experimental runs you can extract:

- the original signal  $\chi(t)$  (from the ancilla, as always),
- the system's energy  $\langle H \rangle$  and magnetisation  $\langle Q \rangle$  (from the garbage alone),
- and — the star of the show — *joint* quantities that combine ancilla and garbage, which normally would require a separate, modified experiment for every quantity you want.

One experiment, many answers. The extra cost is not zero — the random measurement settings add bookkeeping, and the shadow estimates are noisier than dedicated measurements would

be — but no *additional circuits or shots* are needed beyond the ones the standard protocol already runs.

## 4. What we actually did

**Built and validated the machinery (Challenges 1-6).** We implemented the circuit, the data parsing, and the three estimators, and validated every piece against exact references before using it. The full experiment was 256,000 simulated measurement shots spread over 64 time points. Highlights:

- The signal  $\chi(t)$  matched the exact curve everywhere, with honest error bars (the measured scatter was  $1.05\times$  what the error bars predicted — meaning the error bars tell the truth).
- Energy from the garbage data: measured  $+0.071 \pm 0.008$  vs. exact  $+0.074$ .
- Magnetisation: measured  $2.271 \pm 0.005$  vs. exact  $2.267$ .
- Errors shrank with more measurements exactly at the textbook rate (slope  $-0.46$  vs. the ideal  $-0.5$ ), which is the signature of an unbiased estimator with no hidden systematic error.

**Extracted the spectrum with labels (Challenges 7-9).** This is the advanced deliverable. The signal  $\chi(t)$  is like a chord made of four notes; each "note" is an energy level of the magnet. We first tried the honest textbook method (a Fourier transform), which — exactly as theory predicts — hears only three of the four notes: the quietest one sits too close to a much louder neighbour. Then we used a sharper method (the *matrix pencil*, borrowed from radar signal processing) that models the signal directly instead of binning it, and recovered **all four energy levels**, each with:

- its energy (accurate to about 0.03, roughly 1% of the full energy range),
- its loudness/weight (accurate to about 0.006),
- and its **magnetisation label** — the quantum number that says which symmetry sector the level lives in. This label is precisely the information that comes from the garbage register and that the standard Hadamard test cannot provide at all.

Every number carries an uncertainty computed by a bootstrap: we re-ran the whole analysis 200 times on noise-perturbed copies of the data and reported the spread. Satisfyingly, the weakest note has the biggest error bar on its label ( $\pm 0.19$ ) and the loudest has the smallest ( $\pm 0.02$ ) — the analysis knows what it doesn't know.

A detail we're proud of: the notebook includes a "tripwire" test that replaces every piece of code that knows the exact answer with a boobytrap that explodes if touched, then re-runs our analysis. It produced identical output — proof the estimator never peeked at the answer sheet.

**Tried it on a real quantum computer — first attempt failed, and the failure told us exactly what to fix.** We sent a small job (4,000 shots) to IBM's `ibm_marrakesh` processor overnight, using a circuit-construction method that turns out to need about 435 two-qubit operations for this problem size. The result came back heavily degraded: the signal survived at only about 18% of its true size — worse than the  $\sim 34\%$  a generic textbook noise model predicts for a circuit this deep. We reported that as-is; it's a real finding about hardware noise, not a success story.

But the circuit didn't need to be that deep. An alternative way of building the same operation needs only about 100 two-qubit operations — roughly  $4\times$  fewer — because at this small system size (3 qubits) it happens to compile more efficiently than the standard method. We reran the identical experiment with that shallower circuit, on the same hardware: the signal survived at **82%**, not 18%. Running it on a better-calibrated device pushed that to **88%** — close to a clean simulator result. So the real lesson wasn't "this doesn't work on hardware," it was "this doesn't work on a needlessly deep circuit." Two honest caveats. First, each of these is a *single run*, and hardware genuinely varies — an independent run of one identical circuit half an hour later differed by more than a factor of two. Second, this shortcut is a quirk of small system sizes and would not hold at scale, so we report it as a robustness finding about *this* benchmark, not a general recipe.

**Bonus (Challenge 10).** The same recorded data also feeds a completely different algorithm — a "Krylov" energy solver that estimates the lowest occupied energy level. It is knowingly ill-conditioned (the notebook treats it as advisory, not graded); our run landed at  $-1.53$  against a true value of  $-1.92$ , within the method's documented instability band, while our main matrix-pencil method got  $-1.918$  from the same data.

## 5. Why it matters

Measurements on quantum computers are expensive — shots cost time and money, and on today's noisy hardware you want every drop of information per shot. This project demonstrates, end to end and with error bars, that a widely used protocol has been leaving information on the table: data that is routinely discarded can label a system's energy levels by symmetry sector, for no additional circuits — at a measured cost of about 7% more statistical noise on the recycled quantity, plus at most two extra single-qubit gates per qubit.

## 6. Honesty notes

- All headline (Part A/B) results are from an ideal (noiseless) simulator. Real hardware runs are reported separately, always as robustness evidence, never as a headline number: our first attempt (deep circuit) did not survive, later attempts (shallow circuit) survived well (82-88%). Hardware never contributes to the reported spectrum or its uncertainties.
- Headline Part A/B numbers come from **one** shared dataset of 256,000 shots. Side studies are declared separately and are *not* all small: a 12-seed reproducibility study (3,072,000 shots,  $12\times$  the headline set), the scaling and noise studies, five method ablations, Tracks B/C/E, and 13 hardware jobs across three devices.
- Uncertainties are bootstrap-based; validation gates were set at  $5\sigma$ .
- We make no claims about outperforming classical computers — a laptop solves this three-spin system instantly. The system is small *on purpose*, so that every estimate could be graded against an exact answer.